

| Question Number | Answer                                                                                                                                                                                                                                                                                                          | Additional Guidance                 | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------|
| 3ai             | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>• (48 hours) allows time for {bacteria to become visible / extract to work} (1)</li> <li>• 30°C {will not allow growth of (human) pathogens / is high enough for adequate growth (in 48 hours)} (1)</li> </ul> | accept extract to diffuse into agar | 2    |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Additional Guidance                                                                                                                                                                                                                       | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 3aii            | <p>An explanation that includes three of the following points.</p> <ul style="list-style-type: none"> <li>• flame {loop / spreader / neck of culture bottle} to {kill the bacteria / sterilise it / disinfect it} (1)</li> <li>• have a Bunsen lit to move microorganisms in the air away (1)</li> <li>• lift (receiving) {Petri dish / culture bottle} lid partially to minimise entry of bacteria (1)</li> <li>• use disinfectant on workplace to kill bacteria (1)</li> </ul> | <p>do not accept, goggles, do not accept unqualified {gloves, contamination}</p> <p>accept use sterilised equipment, e.g. pipette, gloves, wash hands before start so no bacteria present</p> <p>accept stop bacteria falling in dish</p> | 3    |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                | Additional Guidance                                                                                                                                                                      | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 3bi             | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>• use of squared paper (1)</li> <li>• detail of what to do with paper to get area (1)</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>• measure {diameter / radius} {using callipers / in more than one direction} (1)</li> <li>• detail of calculation of area from this (1)</li> </ul> | <p>e.g. treatment of part squares<br/>do not accept unqualified count squares</p> <p>accept micrometer and/or graticule<br/>accept different angles</p> <p>e.g. <math>\pi r^2</math></p> | 2    |

| Question Number | Answer                                                    | Additional Guidance                                                                                                             | Mark |
|-----------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------|
| 3bii            | type or species of {extract / plant (oil) / bacteria} (1) | <p>Do not accept concentration of extract</p> <p>do not accept if DV or controlled variable given in addition to correct IV</p> | 1    |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Additional Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Mark |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 3biii           | <p>An answer that includes the following point:</p> <ul style="list-style-type: none"> <li>a calculation to compare two relevant results between different {oils / bacteria} (1)</li> </ul> <p>and any three of the following :</p> <ul style="list-style-type: none"> <li>oil A is a more effective (inhibitor of bacterial growth) than oil B (on both bacteria) (1)</li> <li>both oils are more effective (inhibitors of bacterial growth) on <i>B. subtilis</i> (than on <i>E. coli</i>) (1)</li> <li>oil B (inhibits only <i>B. subtilis</i> / does not inhibit <i>E.coli</i>) / oil A inhibits both bacteria (1)</li> <li>biggest (anti-bacterial) effect is A on <i>B. subtilis</i> / smallest (anti-bacterial) effect is B on <i>E. coli</i> (1)</li> </ul> | <p>penalize the idea that the effect is a stimulus to growth once only.</p> <p>e.g. <i>B. subtilis</i> <math>20.66 \div 11.33 = 1.82</math> x more inhibited by A than B</p> <p>oil A has 9.33 greater zone of inhibition than oil B on <i>B. subtilis</i></p> <p>oil A affects <i>B. s.</i> 2.4 times more than it affects <i>E. c.</i> or reverse calculation</p> <p>many other calculations are possible for the mark</p> <p>accept Z.o.I. bigger for more effective</p> <p>accept reverse argument do not piece together, must be clearly stated</p> | 4    |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                    | Additional Guidance                                                                                                                                   | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 3ci             | <p>An answer that includes any two from the following points:</p> <ul style="list-style-type: none"> <li>• size (of antimicrobial molecule) (1)</li> <li>• {solubility (of antimicrobial molecule) / properties of agar} (1)</li> <li>• concentration (of antimicrobial molecules) (1)</li> <li>• (incubation) temperature (1)</li> </ul> | <p>accept molecules in extract for antimicrobial molecule</p> <p>accept concentration of extract in well, concentration gradient</p> <p>ignore pH</p> | 2    |

| Question Number | Answer                                                                                                                                                                                                                       | Additional Guidance                                                                                                                                                                                     | Mark |
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| 3cii            | <p>An answer that includes the following points.</p> <ul style="list-style-type: none"> <li>• calculation of 10% of highest population (1)</li> <li>• reading of concentration of extract that gives 20 alive (1)</li> </ul> | <p>e.g. <math>200 \times 0.1 = 20</math></p> <p>e.g. <math>4.3 \text{ mg dm}^{-3}</math><br/> accept 4.4 for two marks<br/> accept any value between 4.3 and 4.4 for 2<br/> accept 4.2 for one mark</p> | 2    |